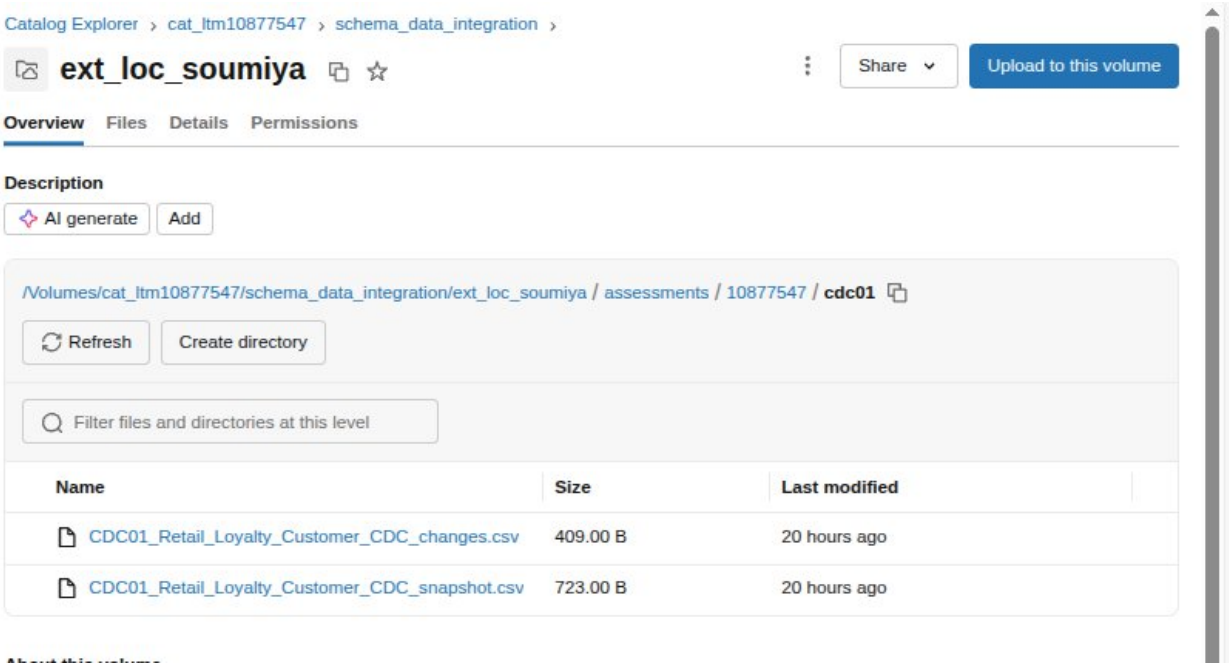
Soumiya Ayyappan_W10_D1

Step 1:



07:12 AM (2s)

```
%sql
use catalog cat_ltm10877547;
create schema if not exists schema_data_integration;
use schema schema_data_integration;
show schemas in cat_ltm10877547;
```

See performance (4)

Table

	databaseName
1	bronze
2	default
3	gold
4	hr
5	information_schema
6	schema_data_integration
7	silver
8	training_assessment_delivery_ops
9	training_assessment_hospitality

07:16 AM (1s) 2

```
from pyspark.sql.functions import *
from pyspark.sql.types import *
schema=StructType([
    StructField("customer_id",StringType(),True),
    StructField("customer_name",StringType(),True),
    StructField("city",StringType(),True),
    StructField("loyalty_tier",StringType(),True),
    StructField("points_balance",IntegerType(),True),
    StructField("is_active",BooleanType(),True),
    StructField("last_updated",TimestampType(),True)
])
df1=spark.read.csv("/Volumes/cat_ltm10877547/schema_data_integration/ext_loc_soumiya/assessments/10877547/cdc01/
CDC01_Retail_Loyalty_Customer_CDC_snapshot.csv",header=True,schema=schema)
```

df1: pyspark.sql.connect.dataframe.DataFrame = [customer_id: string, customer_name: string ... 5 more fields]

07:17 AM (<1s) 3

```
display(df1.count())
```

See performance (1) Checked for improvements

12

07:18 AM (<1s) 4

```
display(df1.filter(col("points_balance").isNull()).count())
```

See performance (1) Checked for improvements

```
1
```

07:19 AM (<1s) 5

```
schema=StructType([
    StructField("event_id",StringType(),True),
    StructField("customer_id",StringType(),True),
    StructField("customer_name",StringType(),True),
    StructField("city",StringType(),True),
    StructField("loyalty_tier",StringType(),True),
    StructField("points_balance",IntegerType(),True),
    StructField("is_active",BooleanType(),True),
    StructField("last_updated",TimestampType(),True),
    StructField("operation",StringType(),True),
])
df2=spark.read.csv("/Volumes/cat_ltm10877547/schema_data_integration/ext_loc_soumiya/assessments/10877547/cdc01/
CDC01_Retail_Loyalty_Customer_CDC_changes.csv",header=True,schema=schema)
```

df2: pyspark.sql.connect.dataframe.DataFrame = [event_id: string, customer_id: string ... 7 more fields]

+ Code + Text Genie Code

07:19 AM (1s) 6

```
display(df2.count())
display(df2.groupBy("operation").count())
```

See performance (2)

```
5
```

Table +

	operation	count
1	I	2
2	U	2
3	D	1

Step 2:

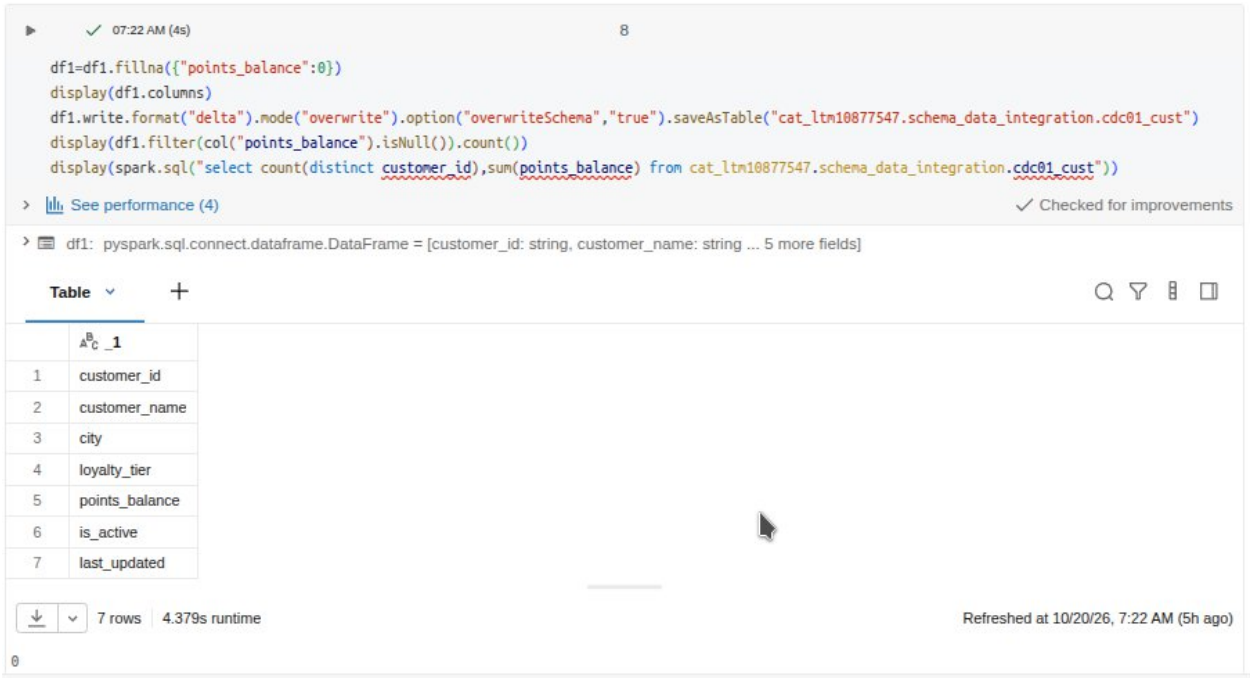

07:22 AM (4s)
8
df1=df1.fillna({"points_balance":0})
display(df1.columns)
df1.write.format("delta").mode("overwrite").option("overwriteSchema","true").saveAsTable("cat_ltm10877547.schema_data_integration.cdc01_cust")
display(df1.filter(col("points_balance").isNull()).count())
display(spark.sql("select count(distinct customer_id),sum(points_balance) from cat_ltm10877547.schema_data_integration.cdc01_cust"))
See performance (4)
Checked for improvements
df1: pyspark.sql.connect.dataframe.DataFrame = [customer_id: string, customer_name: string ... 5 more fields]
Table
_1
1 customer_id
2 customer_name
3 city
4 loyalty_tier
5 points_balance
6 is_active
7 last_updated
7 rows
4.379s runtime
Refreshed at 10/20/26, 7:22 AM (5h ago)
0


Table
count(DISTINCTcustomer_id)
sum(points_balance)
1
12
2560

Step 3:

07:25 AM (2s) 10

```
df2.write.format("delta").mode("overwrite").option("overwriteSchema","true").saveAsTable("cat_ltm10877547.schema_data_integration.cdc01_events")
```

See performance (1) Checked for improvements

07:28 AM (4s) 11

```
display(spark.sql("""merge into cat_ltm10877547.schema_data_integration.cdc01_cust as target
                  using cat_ltm10877547.schema_data_integration.cdc01_events as source on target.customer_id=source.customer_id
                  when matched and source.operation='D' then delete when matched and source.operation='U' then update
                  set target.customer_id=source.customer_id,
                  target.customer_name=source.customer_name,
                  target.city=source.city,
                  target.loyalty_tier=source.loyalty_tier,
                  target.points_balance=source.points_balance,
                  target.is_active=source.is_active,
                  target.last_updated=source.last_updated
                  when not matched and operation='I' then insert
                  (customer_id,customer_name,city,loyalty_tier,points_balance,is_active,last_updated) values(source.customer_id,source.
                  customer_name,source.city,source.loyalty_tier,source.points_balance,source.is_active,source.last_updated);"""))
```

See performance (2) Checked for improvements

Table +

Table +

	num_affected_rows	num_updated_rows	num_deleted_rows	num_inserted_rows
1	5	2	1	2

1 row | 3.738s runtime

Refreshed at 10/20/26, 7:28 AM (4h ago)

10:53 AM (9s) 12

```
display(spark.sql("select * from cat_ltm10877547.schema_data_integration.cdc01_cust"))
```

See performance (1) Checked for improvements

Table +

	customer_id	customer_name	city	loyalty_tier	points_balance	is_active	last_updated
1	C001	Anita	Mumbai	SILVER	120	true	2026-10-01T09:00:00.0
2	C003	Charu	Delhi	SILVER	80	true	2026-10-01T09:00:00.0

13 rows | 8.687s runtime

Refreshed at 10/20/26, 10:53 AM (1h ago)

10:58 AM (1s) 13

```
%sql

SELECT
    customer_id,
    loyalty_tier,
    points_balance,
    is_active,
    last_updated
FROM cat_ltm10877547.schema_data_integration.cdc01_cust
WHERE customer_id IN (
    'C002', 'C004', 'C006', 'C013', 'C014'
)
ORDER BY customer_id;
```

See performance (1) Checked for improvements

	customer_id	loyalty_tier	points_balance	is_active	last_updated
1	C002	PLATINUM	520	true	2026-10-02T10:00:00.000+00:...
2	C006	SILVER	190	true	2026-10-02T10:02:00.000+00:...
3	C013	BRONZE	70	true	2026-10-02T10:05:00.000+00:...
4	C014	SILVER	110	true	2026-10-02T10:08:00.000+00:...

4 rows | 1.056s runtime

Refreshed at 10/20

Step 4:

11:07 AM (1s) 14

```
%sql
--Query final row count, distinct customer IDs, NULL points count, total points, and a tier-level count/SUM breakdown.
select count(*),count(distinct customer_id),sum(case when points_balance is null then 1 else 0 end),sum(points_balance) from cat_ltm10877547.
schema_data_integration.cdc01_cust;
```

See performance (1) Checked for improvements

Table

	count(*)	count(DISTINCTcustomer_id)	sum(CASEWHENpoints_balanceISNULLTHEN1ELSE0END)	sum(points_balance)
1	13	13	0	3080

✓ 11:08 AM (1s) 15

```
%sql
describe history cat_ltm10877547.schema_data_integration.cdc01_cust;
```

See performance (1) ✓ Checked for improvements

Table +

	version	timestamp	userId	userName	operation
1	2	2026-10-20T07:28:49.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	OPTIMIZE
2	1	2026-10-20T07:28:47.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	MERGE
3	0	2026-10-20T07:22:49.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	CREATE OR REPLACE TABLE AS SELECT

✓ 11:11 AM (1s) 16

```
%sql
select * from cat_ltm10877547.schema_data_integration.cdc01_cust version as of 2;
```

See performance (1) ✓ Checked for improvements

Table +

	customer_id	customer_name	city	loyalty_tier	points_balance	is_active	last_updated
1	C001	Anita	Mumbai	SILVER	120	true	2026-10-01T09:00:00.0
2	C003	Charu	Delhi	SILVER	80	true	2026-10-01T09:00:00.0
3	C005	Esha	Mumbai	GOLD	450	true	2026-10-01T09:00:00.0
4	C007	Gita	Pune	BRONZE	40	true	2026-10-01T09:00:00.0
5	C008	Hari	Hyderabad	SILVER	210	true	2026-10-01T09:00:00.0
6	C009	Isha	Mumbai	GOLD	600	true	2026-10-01T09:00:00.0
7	C010	Jai	Chennai	SILVER	160	false	2026-10-01T09:00:00.0
8	C011	Kiran	Delhi	BRONZE	30	true	2026-10-01T09:00:00.0
9	C012	Lata	Hyderabad	GOLD	500	true	2026-10-01T09:00:00.0
10	C002	Bharat	Pune	PLATINUM	520	true	2026-10-02T10:00:00.0
11	C006	Farhan	Delhi	SILVER	190	true	2026-10-02T10:02:00.0

✓ 11:12 AM (1s) 17

```
%sql
select * from cat_ltm10877547.schema_data_integration.cdc01_cust version as of 0;
```

See performance (1) ✓ Checked for improvements

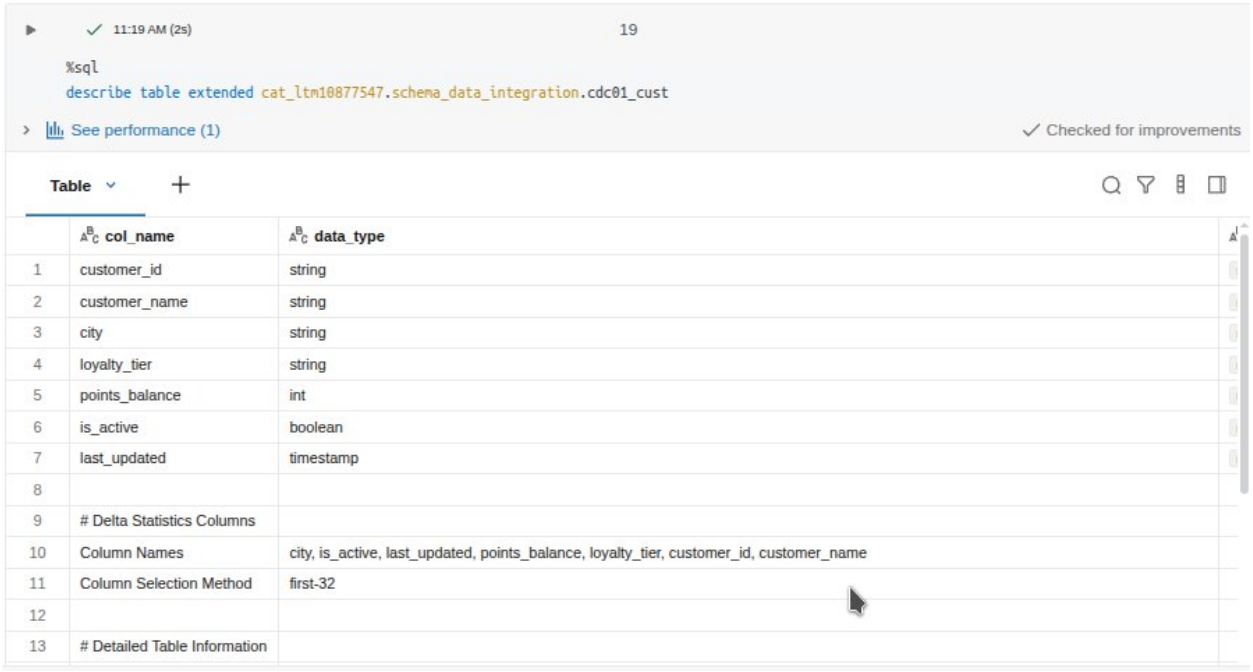
Table +

	customer_id	customer_name	city	loyalty_tier	points_balance	is_active	last_updated
1	C001	Anita	Mumbai	SILVER	120	true	2026-10-01T09:00:00.0
2	C002	Bharat	Pune	GOLD	320	true	2026-10-01T09:00:00.0
3	C003	Charu	Delhi	SILVER	80	true	2026-10-01T09:00:00.0
4	C004	Dev	Chennai	BRONZE	50	true	2026-10-01T09:00:00.0
5	C005	Esha	Mumbai	GOLD	450	true	2026-10-01T09:00:00.0
6	C006	Farhan	Delhi	SILVER	0	true	2026-10-01T09:00:00.0
7	C007	Gita	Pune	BRONZE	40	true	2026-10-01T09:00:00.0
8	C008	Hari	Hyderabad	SILVER	210	true	2026-10-01T09:00:00.0
9	C009	Isha	Mumbai	GOLD	600	true	2026-10-01T09:00:00.0
10	C010	Jai	Chennai	SILVER	160	false	2026-10-01T09:00:00.0
11	C011	Kiran	Delhi	BRONZE	30	true	2026-10-01T09:00:00.0
12	C012	Lata	Hyderabad	GOLD	500	true	2026-10-01T09:00:00.0

Step 5:



	format	id	name	description	location
1	delta	f16a55f4-ae6e-4cfa-9dff-16a23454c97d	cat_ltm10877547.schema_data_integration.cdc01_cust	null	abfss://unity-catalog-storag



	col_name	data_type
1	customer_id	string
2	customer_name	string
3	city	string
4	loyalty_tier	string
5	points_balance	int
6	is_active	boolean
7	last_updated	timestamp
8		
9	# Delta Statistics Columns	
10	Column Names	city, is_active, last_updated, points_balance, loyalty_tier, customer_id, customer_name
11	Column Selection Method	first-32
12		
13	# Detailed Table Information	

11:21 AM (1s) 20

```
%sql
SELECT loyalty_tier, COUNT(*) AS customers, SUM(points_balance) AS total_points
FROM cat_ltm10877547.schema_data_integration.cdc01_cust
GROUP BY loyalty_tier ORDER BY loyalty_tier;
```

See performance (1)

Table

	loyalty_tier	customers	total_points
1	BRONZE	3	140
2	GOLD	3	1550
3	PLATINUM	1	520
4	SILVER	6	870